

Tutorial on Moy-Prasad filtrations, Yu construction and characters

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Lecture 1. F a local field, $F \supset \mathcal{O}_F \supset \mathfrak{m}_F$, $k = \mathcal{O}_F / \mathfrak{m}_F$

G conn. red. gp / F , C coeff. field, any alg closed field with
 $\text{char } C \neq p = \text{char } k$.

Let $(\pi, V) \in \text{Rep}_C(G(F))$. $\pi: G(F) \rightarrow \text{Aut}_C(V)$ smooth s.t. every $v \in V$ has
an open stabilizer.

$\mathcal{H}(G(F)) := C_c^\infty(G(F))$ dg space of compactly supp. locally constant C -valued
measures.

$\mathcal{H}(G(F))$ assoc. C -alg. by convolution.

For $\mu \in \mathcal{H}(G(F))$, we put $\pi(\mu) = \int_G \pi(g) \mu(g) \in \text{End}_C(V)$.

Th. (Jacquet, Bernstein, Vignéras) Suppose π has finite length, then $\forall K \subset G(F)$
open compact subgroup, $\dim_C \pi^K < +\infty \Rightarrow \text{im}(\pi(\mu)) \subset \pi^K$ is finite dim'l

In this case, we define $\Theta_\pi: \mathcal{H}(G(F)) \rightarrow C$, $\mu \mapsto \text{Tr}(\pi(\mu))$.

the character of π .

Th. (Harish-Chandra) Suppose π has finite length and $\text{char}(F) = 0$, then

(1) $\Theta_\pi|_{\substack{C_c^\infty(F) \\ \cap \text{open dense} \\ G(F)}}$ is locally constant.

i.e. $\exists F_\pi \in (C^\infty G^{\text{rs}}(F))$ C -valued locally constant function on $G^{\text{rs}}(F)$.

s.t. if $\mu \in \mathcal{H}(G(F))$ is supported on $G^{\text{rs}}(F)$, then

$$\Theta_\pi(\mu) = \text{Tr}(\pi(\mu)) = \int_{G^{\text{rs}}(F)} F_\pi(g) \mu(g).$$

(2) Moreover, if $\text{char}(C) = 0$, then Θ_π is "determined by F_π (canonically)"
(local character expansion)

Moreover, if $C = \mathbb{C}$, then $\Theta_\pi^{(\mu)} = \text{Tr}(\pi(\mu)) = \int_{G^{\text{rs}}(F)} F_\pi(g) \mu(g)$

$\forall \mu \in \mathcal{H}(G(F))$ where the RHS is defined by absolute convergence.

§2. Moy-Prasad filtration.

C any alg. closed field of $\text{char} \neq p$.

Th. (Moy-Prasad, Vignéras) $\text{Rep}_C(G(F)) = \prod_{r \in \mathbb{Q}_{\geq 0}} \text{Rep}_C(G(F))^{\text{depth} \leq r}$

Firstly, assume G split / F , let $\mathcal{G}$ be an integral model of G , i.e.

Smooth gp scheme / $\mathcal{O}_F$ with conn. reductive geom. fibre and $\mathcal{G} \times_{\mathcal{O}_F} F = G$.

Write for $r \in \mathbb{R}_{\geq 0}$ that

$$G(F)_r := \ker \left(\mathcal{G}(\mathcal{O}_F) \rightarrow \mathcal{G}(\mathcal{O}_F / \mathfrak{m}_F^{r+1}) \right) \triangleleft \mathcal{G}(\mathcal{O}_F) = G(F)_0.$$

$\forall r \in \mathbb{R}$, Lie $G(F)_r = \mathfrak{m}_F^{r+1} \text{ Lie } \mathcal{G}(\mathcal{O}_F)$ depend on $\mathcal{G}$.

Suppose $S \subset G$ is a F -split max'l torus. Its closure $\bar{S} \subset G$ is an $\mathcal{O}_F$ -split max'l torus.

$$\text{Write } S(F)_r = G(F)_r \cap S(F)$$

$$\forall \alpha \in \Phi(G, S), \quad G_\alpha(F)_r = G(F)_r \cap \underbrace{G_\alpha(F)}_{\substack{\text{root subgp} \\ S \\ F}}$$

Fact $G(F)_r = \langle S(F)_r, G_\alpha(F)_r : \alpha \in \Phi(G, S) \rangle$

Let $\lambda \in X_*(S) \otimes \mathbb{R}$. With $x = (g, s, \lambda)$ we define Moy-Prasad subgp

$$G(F)_{x,r} = \langle S(F)_r, G_\alpha(F)_{r-\langle \alpha, \lambda \rangle} : \alpha \in \Phi(G, S) \rangle$$

Def. Gp of the form $G(F)_{x,0}$ is called a parahoric subgp.

Def. $\mathcal{B}(G, F) = \{ (g, s, \lambda) \} / \sim$ where $(g, s, \lambda) \sim (g', s', \lambda')$

$$\text{if } G(F)_{(g,s,\lambda),r} = G(F)_{(g',s',\lambda'),r}, \quad \forall r \in \mathbb{R}_{\geq 0}.$$

Suppose G splits $|E$, $E|F$ finite tame Galois ext'n.

Then $\text{Gal}(E|F) \curvearrowright \mathcal{B}(G, E)$, may define $\mathcal{B}(G, F) := \mathcal{B}(G, E)^{\text{Gal}(E|F)}$

$$\forall x \in \mathcal{B}(G, F) \text{ and } r > 0, \quad G(F)_{x,r} := G(E)_{x,r}^{\text{Gal}(E|F)} = G(E)_{x,r} \cap G(F).$$

Rem. $G(F)_{x,0} \subsetneq G(E)_{x,0} \cap G(F)$ in general.

(= if G is semisimple simply conn'd)

Def. $G(F)_{x,r+} = \varinjlim_{s \rightarrow r+} G(F)_{x,s}$

Ex $g = GL_2$, $\mathfrak{g} = \begin{pmatrix} * & * \\ & * \end{pmatrix}$, $\lambda = \frac{1}{4}(e_1 - e_2) = \frac{1}{4}\alpha^\vee$.

$$x = (g, s, \lambda), \quad u(F)_{x,r} = \begin{cases} \begin{pmatrix} \mathcal{O}_F^\times & \mathcal{O}_F \\ m_F & \mathcal{O}_F^\times \end{pmatrix}, & r=0 \\ \begin{pmatrix} 1+m_F & \mathcal{O}_F \\ m_F & 1+m_F \end{pmatrix}, & r \in (0, \frac{1}{2}] \\ \begin{pmatrix} 1+m_F & m_F \\ m_F^2 & 1+m_F \end{pmatrix}, & r \in (\frac{1}{2}, 1] \end{cases}$$

Def. A rep'n $\pi \in \text{Rep}_C(G(F))$ has depth $r \in \mathbb{R}_{\geq 0}$ ($\mathbb{Q}_{\geq 0}$) if

(i) π is generated by $\pi^{u(F)_{x,r}}$, when x runs through $\mathcal{B}(G, F)$

(ii) If $r > 0$, $\pi^{u(F)_{x,r}} = 0$, $\forall x \in \mathcal{B}(G, F)$.

$$\text{Rep}_C(G(F)) = \prod_{r \in \mathbb{Q}_{\geq 0}} \text{Rep}_C(G(F))^{\text{depth} \leq r}$$

Ex. $\pi \in \text{Rep}_C(GL_2(F))$ has depth $1/2$ if π is gen'd by $\pi^{\begin{pmatrix} 1+m_F & m_F \\ m_F^2 & 1+m_F \end{pmatrix}}$

and $\begin{pmatrix} \mathcal{O}_F/m_F & \\ m_F/m_F^2 & \end{pmatrix} \sim \pi^{\begin{pmatrix} 1+m_F & m_F \\ m_F^2 & 1+m_F \end{pmatrix}}$ is annihilated by

abelian (always when $r > 0$)

$$\begin{pmatrix} 1+m_F & m_F \\ m_F & 1+m_F \end{pmatrix} \text{ and } \begin{pmatrix} 1+m_F & \mathcal{O}_F \\ m_F^2 & 1+m_F \end{pmatrix}$$

$\parallel$
 $u(F)_{ot}$

$\parallel$
 $u(F)_{y,ot}$

$\parallel$
 $(g, s, \frac{1}{2}\alpha^\vee)$

Lecture 2. depth-zero supercuspidal

F a local field w res field k . C alg closed field of char $\neq 0$.

pages

Suppose (1) $x \in B(G, F)$ is a vertex, i.e. $G(F)_{x,0} \subset G(F)_{y,0} \Rightarrow x=y$.

Have $\text{Stab}_{G(F)}(x) \supset G(F)_{x,0} \supset G(F)_{x,0+}$

(2) $\sigma \in \text{Irr}_C(\text{Stab}_{G(F)}(x))$ is trivial on $G(F)_{x,0+}$.

s.t. $\sigma|_{G(F)_{x,0}/G(F)_{x,0+}}$ is a $\oplus$ of cuspidal reps.

$\{ \}$
 $G_x(k)$, G_x conn. red. gp / k .

$$\Leftrightarrow \sigma|_{G(F)_{y,0+} \cap G(F)_{x,0}} = 0$$

$\forall y \in B(G, F), y \neq x$.

Th. (Moy-Prasad)

$\sigma|_{\text{c-ind}_{\text{Stab}_{G(F)}(x)}^{G(F)}} \sigma$ is irred $\swarrow$ supercuspidal. and all irred depth-0 supercuspidal
 depth 0

arise this way. for some (x, σ) .



Υ_u (tame) supercuspidal

Assume G splits / E , E/F finite Galois tame.

$p = \text{char}(k) \nmid |W_G|$ (absolute Weyl gp)

Very roughly, we'd like $K_{\Upsilon_u}^{\text{open}} \subset G(F)$ and $P_{\Upsilon_u} \in \text{Irr}_C(K_{\Upsilon_u})$ s.t.

Th. (Yu, Kim, Fintzen) $\text{c-ind}_{K_{\Upsilon_u}}^{G(F)} P_{\Upsilon_u}$ is an irred supercuspidal, and all irred

Supercusp. arise this way

Input $(a^0 \subset a^1 \subset a^2 \subset \dots \subset a^d = a)$ of closed F -subgp ^{s.t. (a)} by enlarging E if necc.,

$T \subset a^0/E \subset a^1/E \subset \dots \subset a^d/E$ are Levi subgps. (b) $Z(a^0)/Z(a)$ anisotropic / F .

$$(2) \quad x \in B(\mathcal{A}^0, F) \hookrightarrow B(\mathcal{A}^1, F) \hookrightarrow \dots \hookrightarrow B(\mathcal{A}^d, F)$$

$$(i) \quad F = k((t)), \quad (\mathcal{A}^0 \subset \mathcal{A}^1 \subset \mathcal{A}^2) = \mathcal{A}L_1^4 \subset \mathcal{A}L_2^2 \subset \mathcal{A}L_4 \quad / k$$

base change from k to $\mathcal{O}_F$, they give such x , but $\mathbb{Z}(\mathcal{A}^0)/\mathbb{Z}(\mathcal{A})$ is not anisotropic.

$$(ii) \quad k_4 | k_2 | k \text{ quad. ext'n.} \quad \mathcal{A}^0 = \text{Res}_k^{k_4} \mathcal{A}L_1 \subset \mathcal{A}^1 = \text{Res}_k^{k_2} \mathcal{A}L_2 \subset \mathcal{A}^2 = \mathcal{A}L_4$$

$$(3) \quad 0 < \gamma_0 < \gamma_1 < \dots < \gamma_{d-1} \quad \text{rat'l no.}$$

$$K_{\gamma_u} = \text{Stab}_{\mathcal{A}^0(F)}(x) \cdot \prod_{i=1}^d \mathcal{A}^i(F)_{x, \frac{\gamma_i-1}{2}}$$

More data for P_{γ_u} : (4) homomorphism

$$\phi_i : \mathcal{A}^i(F) \rightarrow C^* \quad \text{for } i=0, 1, \dots, d-1 \quad \text{s.t.} \quad \text{they are } \mathcal{A}^{i+1} \text{-generic of depth } \gamma_{i-1}''$$

$$(a) \quad \phi_i|_{\mathcal{A}^i(F)_{x, \gamma_{i-1}''}} \equiv 1$$

$$(b) \quad \phi_i \left(N_{E/F} \left(x^\vee (1 + m_E^{(\gamma_{i-1})} e(E|F)) \right) \right) \neq 1, \quad \forall x^\vee \in \mathbb{F}^\vee(\mathcal{A}^{i+1}, T) \setminus \mathbb{F}^\vee(\mathcal{A}^i, T)$$

$$(5) \quad \sigma \in \text{Irr}_C(\text{Stab}_{\mathcal{A}^0(F)}(x)), \quad \text{trivial on } \mathcal{A}^0(F)_{x,0+} \quad \text{and } \sigma|_{\mathcal{A}^0(F)_{x,0}/\mathcal{A}^0(F)_{x,0+}}$$

is a $\oplus$ of cuspidals.

$$\rightsquigarrow P_{\gamma_u} \in \text{Irr}_C(K_{\gamma_u})$$

$$\text{ind}_k^{K_{\gamma_u}} \hat{\phi}_0 \otimes \hat{\phi}_1$$

Work over E , Choose a parabolic subgroup $P \subset G$ having $\mathcal{L}^\circ$ as Levi

$$\text{Consider } K_{Y_u}^+ = \text{Stab}_{\mathcal{L}^\circ(F)}(x) \left(\prod_{i=1}^d \mathcal{L}^i(F)_{x, \frac{r_i-1}{2}+} \right)$$

$$K_{Y_u}^L = \text{Stab}_{\mathcal{L}^\circ(E)}(x) \left(\prod_{i=1}^d \mathcal{L}^i(F)_{x, \frac{r_i-1}{2}+} \right) \prod \mathcal{L}^i(E)_{x, \frac{r_i-1}{2}} \cap P(E)$$

$$\hat{\phi}_0 : K_{Y_u}^L(E) \rightarrow \mathbb{C}^\times \text{ by } \hat{\phi}_0|_{K_{Y_u}^L(E) \cap \mathcal{L}^\circ(E)} = \phi_0$$

$$\hat{\phi}_0|_{K_{Y_u}^L \cap \text{root subgroup}} \equiv 1$$

$$\hat{\phi}_1 : K_{Y_u}^L(E) \rightarrow \mathbb{C}^\times \text{ by } \hat{\phi}_1|_{K_{Y_u}^L(E) \cap \mathcal{L}^1(E)} = \phi_1$$

$$\hat{\phi}_1|_{K_{Y_u}^L(E) \cap (\text{root subgroup outside } \mathcal{L})} = 1$$

$$\rho_{Y_u} = \text{ind}_{K_{Y_u}^L(E)}^{K_{Y_u}(E)} \hat{\phi}_0 \otimes \hat{\phi}_1 \in \text{Irr}_{\mathbb{C}}(K_{Y_u})$$



$$g \cdot h \in \underbrace{\mathcal{L}^i(F)_{x, \frac{r_i-1}{2}} / \mathcal{L}^i(F)_{x, \frac{r_i-1}{2}+} \left(\mathcal{L}^{i-1}(F) \cap \mathcal{L}^i(F)_{x, r_i} \right)}$$

st. $\phi_i(g h g^{-1} h^{-1})$ defines a symplectic form on S

$$\uparrow \\ \mu_P(\mathbb{C}^\times)$$

1. The first part of the paper is devoted to the study of the

properties of the function $f(x)$ defined by the equation

$f(x) = \int_0^x f(t) dt$ and to the study of the function

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